

Overview of metabolism in the fed and fasting state

Dr. Sharmila Upadhyia

shupadhyia@sgu.edu

Professor and Chair, Department of Biochemistry

St. Georges University

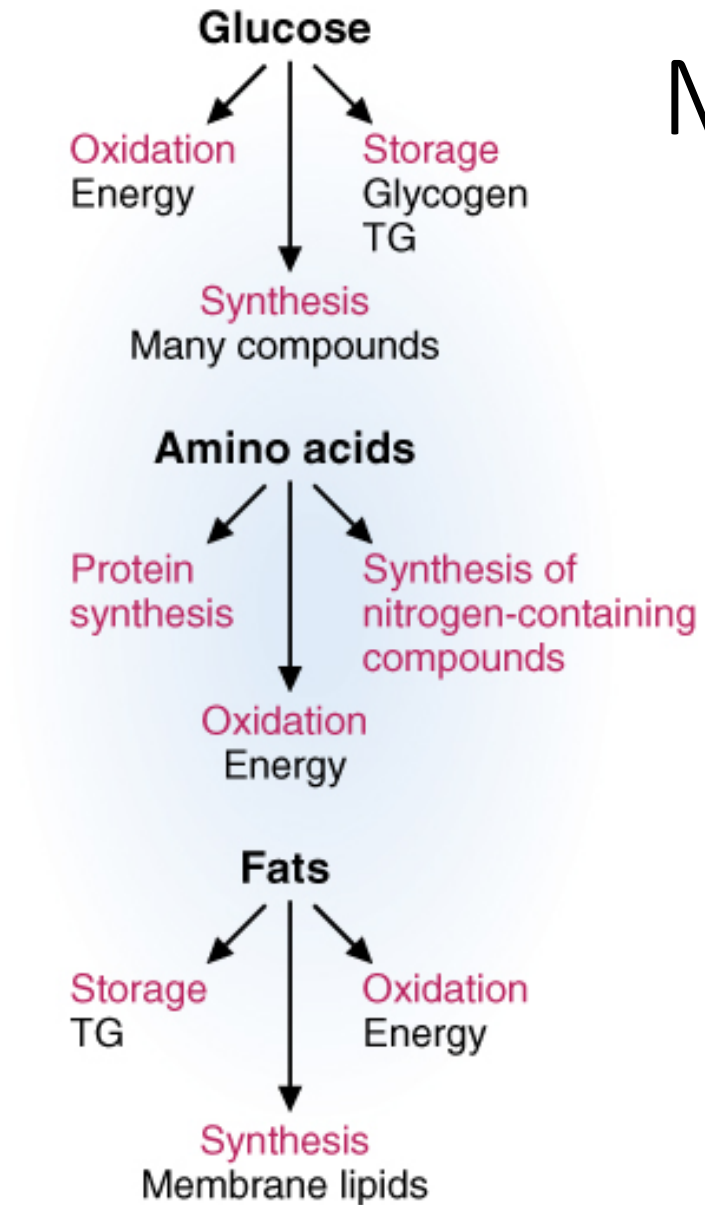
This DLA has two videos and provides an overview of metabolism in the fed and fasting state

Objectives: DLA: Overview of metabolism in the fed fasting state

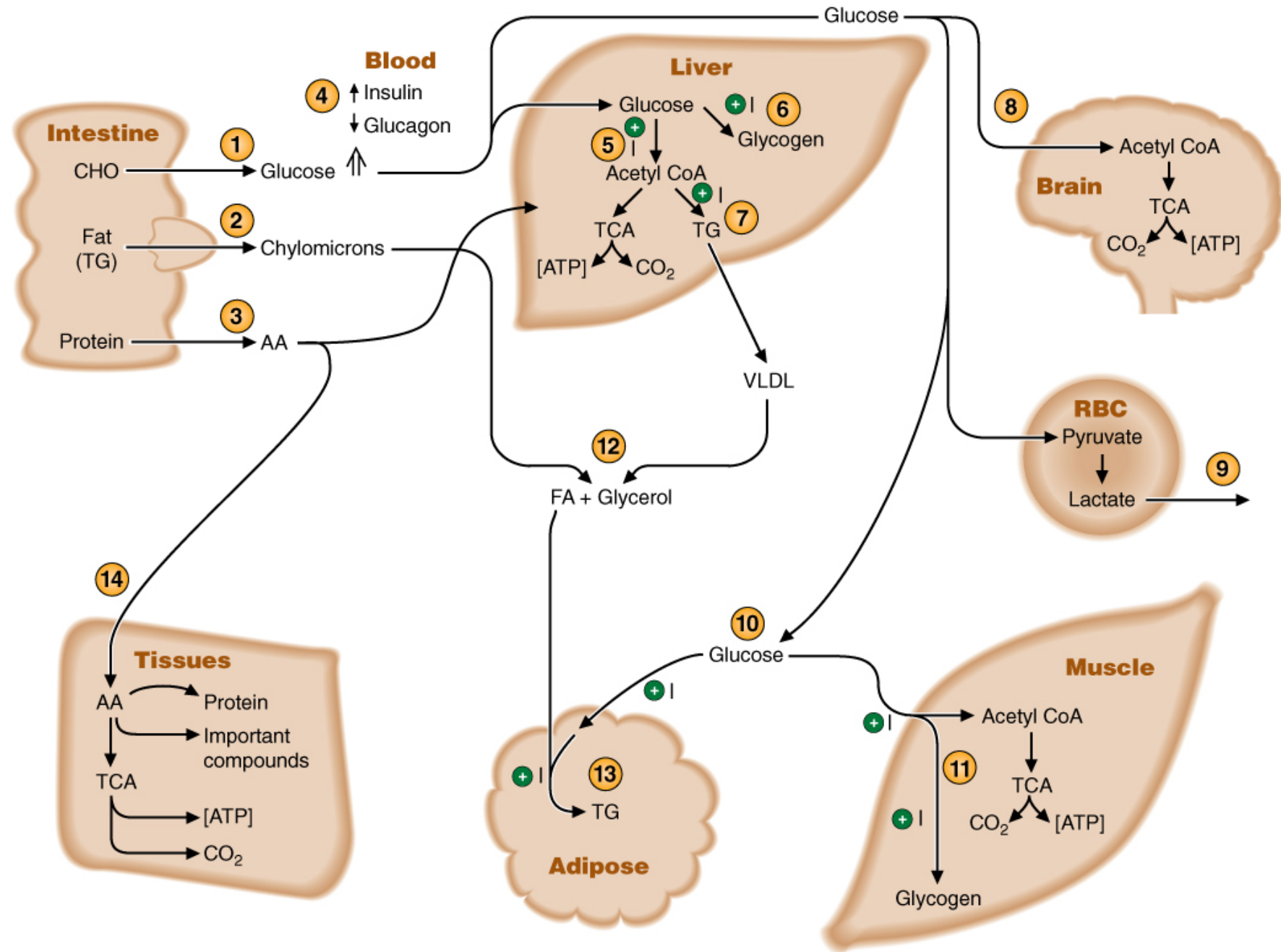
- After DM Lecture 01

SOM. MK.III. BPM2.3. DM.2.BCHM. 1106	Outline the metabolic changes that occur during the feed/fast cycle.
SOM. MK.III. BPM2.3. DM.2.BCHM. 1107	Discuss pathways that are active/inactive in each major organ/tissue during the feed/fast cycle and their regulation and coordination.

Major fates of fuel in fed state



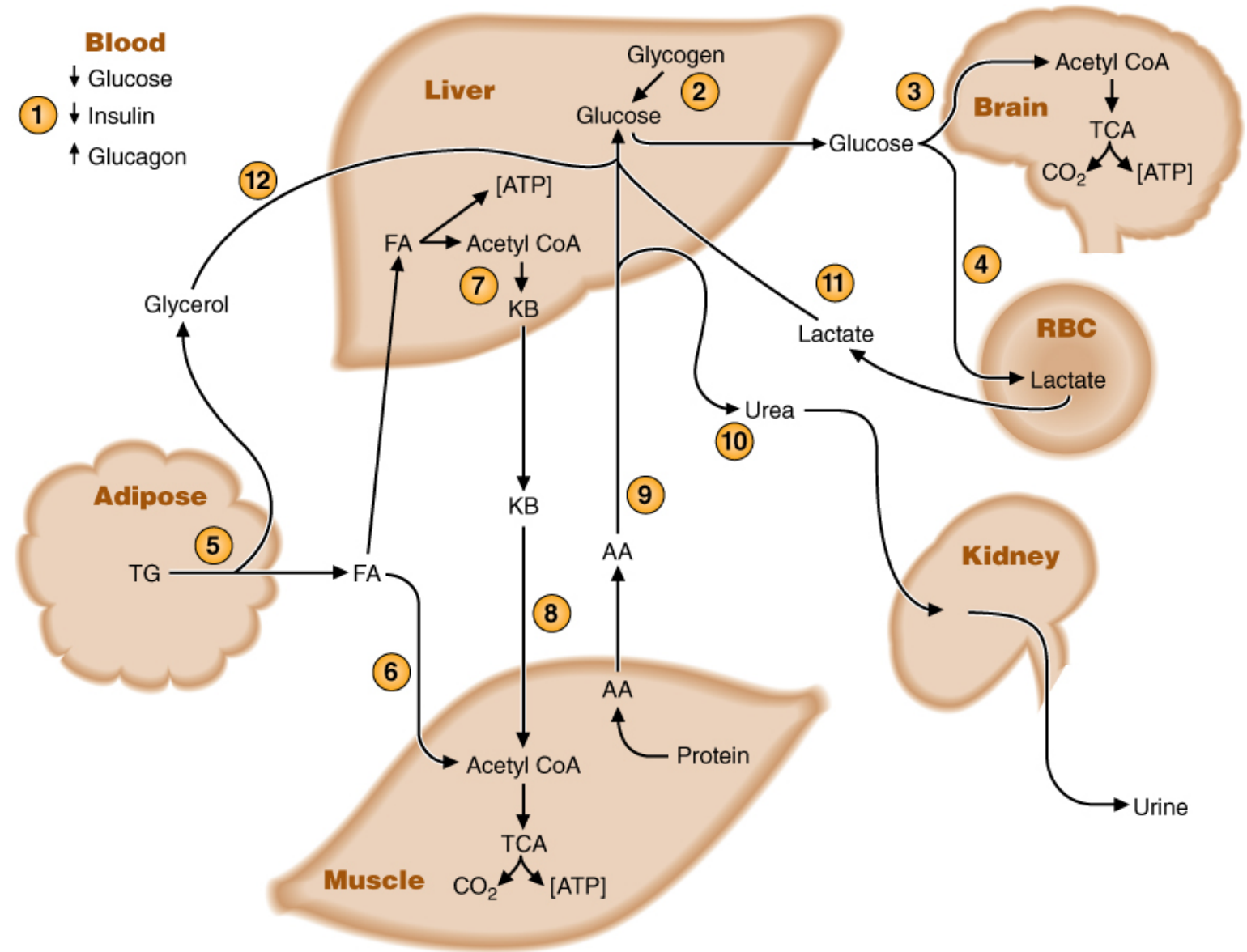
Metabolism in Fed state



Metabolism in Fed state (refer to numbers on previous slide)

1. Absorption of dietary glucose increases plasma glucose
2. Dietary TAG are absorbed and transported as chylomicrons
3. Dietary protein is absorbed as amino acids
4. Increased plasma glucose results in insulin secretion and decreases glucagon secretion (Increases insulin: glucagon ratio)
5. Liver glycolysis is stimulated
6. Liver glycogenesis is favored
7. Fatty acid synthesis from acetyl CoA is stimulated and triacylglycerol (TAG) formation is stimulated
8. Brain uses plasma glucose as an energy source (complete oxidation to CO₂ and H₂O)
9. RBCs uses glucose as an energy source to form lactate which is transported to liver by Cori cycle
10. Glucose entry into adipocytes is favored due to presence of GLUT-4 on adipocyte membrane
11. Glucose entry into muscle is favored (GLUT-4) and glycogen synthesis in muscle is stimulated
12. Lipoprotein lipase present in endothelium of blood vessels of adipocytes and muscle is active and forms Fatty acids and glycerol which are taken up by adipocytes.
13. In adipocytes, formation and storage of TAG is favored. Glycerol-phosphate is provided by DHAP from glycolysis.
14. Amino acid entry and uptake into all tissues is favored. Protein synthesis is active

Metabolism 10-12 hours after meal (Overnight fast)



Metabolism 10-12 hours after meal (Overnight fast) (refer to numbers on previous slide)

1. A fall in plasma glucose reduces insulin: glucagon ratio
2. Liver glycogenolysis is stimulated
3. Plasma glucose is used for brain metabolism. Glucose is completely oxidized to CO₂ and H₂O
4. RBCs use glucose and form lactate.
5. Adipose tissue lipolysis is active (TAG broken down to Free fatty acids and glycerol by hormone sensitive lipase)
6. Fatty acids are used for energy by skeletal and cardiac muscle
7. Fatty acid oxidation in the liver provides energy to the liver and forms ketone bodies
8. Ketone bodies are used as sources of energy by the muscle
9. Muscle Proteolysis is active and releases amino acids. The amino acids in liver are used for gluconeogenesis.
10. Ammonia generated by amino acid metabolism is detoxified to Urea which is lost through the kidneys (urine)
11. Cori cycle transports lactate to liver which is used for gluconeogenesis in liver.
12. Glycerol formed in adipose tissue is also used as a precursor of gluconeogenesis